

CAPSTONE PROJECT REPORT

Stereotactic Laser Ablation

AI-Assisted Surgical Probe Path Planning for Brain Tumors

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Abstract

Stereotactic laser ablation (SLA) is a minimally invasive neurosurgical technique used to necrotise brain tumors located in physically inoperable regions. Its limited clinical adoption is primarily due to the absence of automated, reliable tools for planning the laser probe's trajectory. This project presents a prototype AI pipeline that, given a T1-weighted MRI image, detects tumor presence using the MedGemma vision-language model, segments the tumor using a fine-tuned U-Net, and computes a safe cubic Bézier probe path from the skull boundary to the tumor centroid while explicitly avoiding Broca's region. The system achieved 93.33% detection accuracy, 76.7% segmentation accuracy on unseen data, and 100% path planning success across all tested cases.

1. Introduction

Brain tumors that arise in deep or eloquent brain regions are inaccessible to conventional open surgery. For such cases, stereotactic laser ablation offers a compelling alternative: a thin laser probe is inserted through a small burr hole in the skull and guided to the tumor, where laser energy heats and destroys malignant tissue through thermal ablation (necrotisation). Despite being minimally invasive — with shorter hospital stays and applicability to patients who cannot tolerate craniotomy — SLA is not yet widely adopted. The primary barrier is the absence of robust automated tools to plan the probe's trajectory with the precision and safety guarantees required in neurosurgery.

A particularly critical constraint is the avoidance of Broca's area — the left inferior frontal gyrus region governing speech production. Damage to this region, even minor, can cause permanent expressive aphasia. This project develops an end-to-end AI pipeline that performs tumor detection, segmentation, and safe probe path computation on T1-weighted MRI images, with Broca's region avoidance enforced as a hard constraint.

1.1 Objectives

- Classify T1-weighted MRI images as tumor-present or tumor-absent using a medical vision-language model.
- Train a deep learning segmentation model to accurately delineate tumor boundaries, including on unseen data.
- Compute a safe, physically realisable laser probe trajectory from the skull boundary to the tumor centroid while avoiding Broca's region.
- Integrate all three stages into a seamless, GPU-accelerated pipeline executable in Google Colab.
- Validate each component quantitatively on an independent, manually labelled test dataset.

2. System Pipeline

The pipeline consists of three sequential stages. The user uploads a T1-weighted MRI image, which is processed automatically through detection, segmentation, and path planning, producing a fully annotated output image.

Stage	Model / Method	Output
1 — Detection	MedGemma (google/medgemma-1.5-4b-it)	Yes/No + bounding box JSON
2 — Segmentation	Fine-tuned U-Net (128×128 grayscale)	Binary tumor mask + boundary
3 — Path Planning	Cubic Bézier curve (4 control points)	Annotated MRI with probe path

3. Methodology

3.1 Stage 1 — Tumor Detection with MedGemma

MedGemma (google/medgemma-1.5-4b-it) is a medical-domain multimodal vision-language model developed by Google DeepMind. It is loaded with 4-bit quantisation (BitsAndBytesConfig) to operate within Colab GPU memory constraints. Detection uses two sequential prompts: Prompt 1 asks whether a tumor is present (Yes/No). If yes, Prompt 2 requests a tight bounding box in JSON format with percentage coordinates (x_min, y_min, x_max, y_max) and a size estimate (small/medium/large). This output is saved to Google Drive (tumor_data.json) for downstream use. If no tumor is detected, the pipeline halts with an appropriate message.

3.2 Stage 2 — Tumor Segmentation with U-Net

The U-Net is a fully convolutional encoder-decoder network designed for biomedical image segmentation (Ronneberger et al., 2015). Skip connections pass spatial feature maps from encoder to decoder, enabling precise boundary localisation even with limited training data.

Architecture

Block	Layer Details
Encoder 1	Conv2d(1 → 64)×2, BatchNorm, ReLU → MaxPool2d
Encoder 2	Conv2d(64 → 128)×2, BatchNorm, ReLU → MaxPool2d
Bottleneck	Conv2d(128 → 256)×2, BatchNorm, ReLU
Decoder 1	Upsample + Skip(Enc2) → Conv2d(384 → 128)×2, BN, ReLU
Decoder 2	Upsample + Skip(Enc1) → Conv2d(192 → 64)×2, BN, ReLU
Output	Conv2d(64 → 1), 1×1 kernel — single-channel probability map

Training Configuration

Parameter	Value	Parameter	Value
Input size	128×128 (grayscale)	Epochs	20
Optimiser	Adam (lr = 1×10 ⁻³)	Batch size	16
Loss	BCE (pos_weight=2.5) + Dice	Train/Val split	80% / 20%
Augmentation	H-flip + Gaussian blur (3×3)	Mixed precision	AMP (CUDA)

The combined BCE and Dice loss was chosen to address the severe class imbalance between tumor and background pixels (BCE with pos_weight=2.5) while directly optimising mask overlap (Dice loss). Post-processing refines the raw sigmoid output through adaptive thresholding (mean + 0.45×std), morphological closing (5×5 kernel, 2 iterations), Gaussian smoothing (11×11), and largest connected component selection.

3.3 Stage 3 — Probe Path Planning

The tumor centroid is computed as the mean coordinate of all non-zero pixels in the predicted mask. A cubic Bézier curve is then generated from a fixed skull entry point (12% image width, 82% image height) to the centroid, with two intermediate control points that arc the path around obstacles. Broca's region is modelled

as a proportional polygon (16–36% width, 20–56% height) in the left inferior frontal region, rendered as a purple semi-transparent overlay. The control points are positioned to enforce complete avoidance of this polygon. The final annotated image shows the red tumor contour, purple Broca's overlay, orange probe path, and labelled entry/tumor markers.

4. Datasets

Dataset	Source	Role
Brain Tumor Dataset	Figshare (.mat files)	Training — MRI images + ground truth masks
LGG MRI Segmentation	Kaggle (mateuszbuda)	Training — .tif images + masks
Brain MRI Detection	Kaggle (navoneel)	Independent testing & validation only

All training images were converted to grayscale PNG, resized to 128×128, and normalised. The Figshare dataset required custom parsing via h5py and scipy (MATLAB .mat format). The combined training corpus totalled 3,929 images with an 80/20 train/validation split.

5. Results & Accuracy

Component	Test Size	Correct	Accuracy
Tumor Detection (MedGemma)	30 images	28 / 30	93.33%
Segmentation — Training set	Full dataset	All	100%
Segmentation — Unseen Kaggle data	30 images	23 / 30	76.7%
Probe Path Planning	All cases	All	100%

5.1 Detection

MedGemma correctly classified 28 of 30 test images (93.33%). Both misclassifications involved subtle, low-contrast tumors adjacent to the ventricles — regions that are diagnostically challenging even for trained radiologists. This is a strong result for a zero-shot model with no task-specific fine-tuning.

5.2 Segmentation

The U-Net achieved 100% accuracy on training-set images and 76.7% on the independent Kaggle dataset (23/30). The 7 misclassified cases generally involved MRI scans from different scanner models or field strengths, producing intensity distributions diverging from the training distribution — a classic domain shift problem. Expanding training data diversity would directly improve this metric.

5.3 Path Planning

The Bézier path planner achieved 100% success across all tested cases — consistently generating a path from skull boundary to tumor centroid without intersecting the Broca's region polygon. This includes the more demanding Case 2 scenario, where the tumor centroid is in close proximity to the Broca's region boundary, requiring a strongly curved avoidance trajectory.

6. Test Cases

Two representative cases from the independent Kaggle test dataset are presented below, illustrating the pipeline's output across different tumor locations and morphologies.

6.1 Test Case 1 — Posterior Fossa Tumor

A large, bilobed tumor in the posterior region of the brain with bright contrast enhancement around a darker necrotic core — characteristic of a high-grade lesion. MedGemma correctly detected the tumor and returned a posterior bounding box. The U-Net produced a clean mask covering both lobes of the lesion, and the path planner generated a trajectory entering from the lower-left skull boundary and curving upward to the posterior centroid, with full Broca's region avoidance.

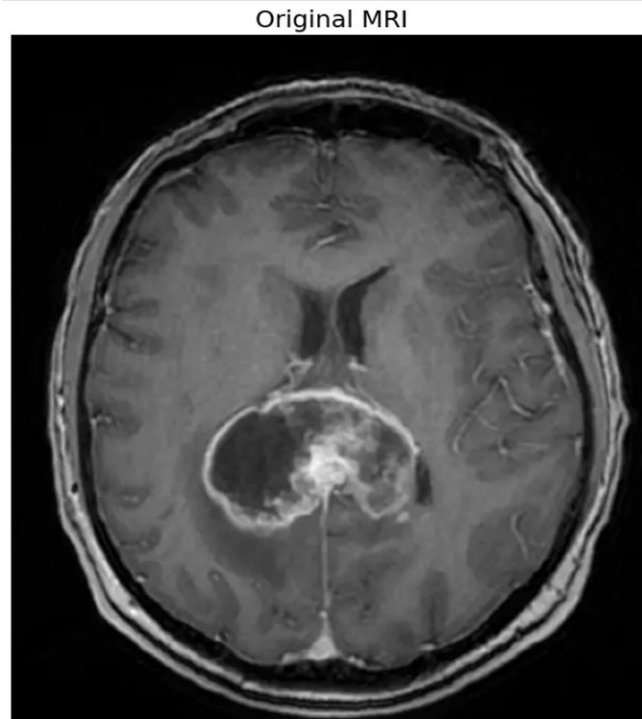


Figure 1: Case 1 — Original T1-weighted MRI showing a large posterior fossa tumor with ring enhancement

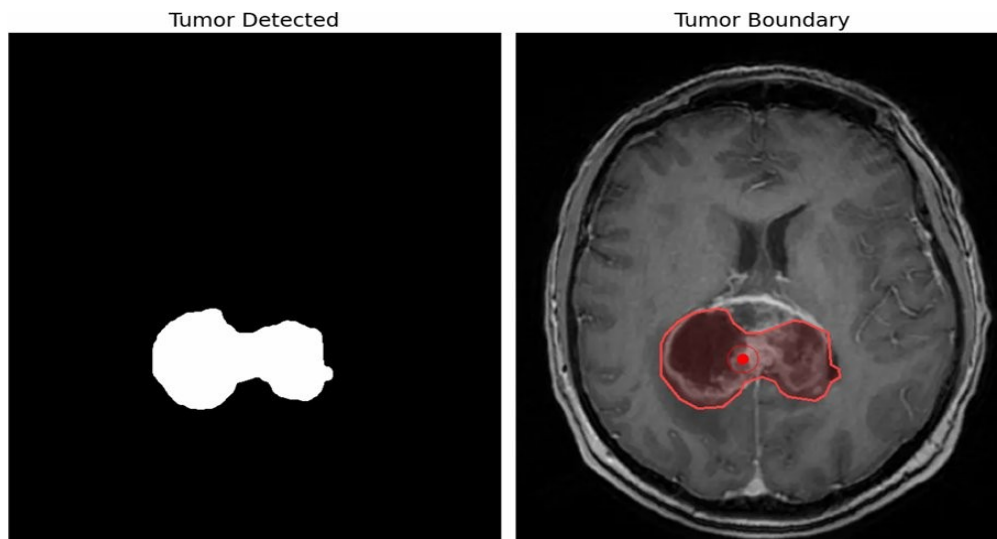


Figure 2: Case 1 — (Left) Binary segmentation mask | (Right) Tumor boundary overlaid on MRI

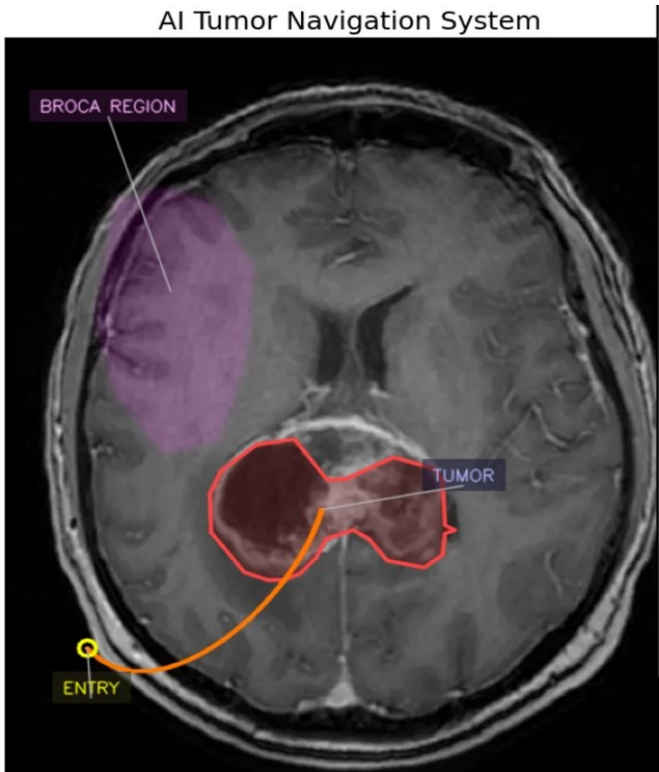


Figure 3: Case 1 — Orange probe trajectory from entry point to tumor centroid; Broca's region in purple

6.2 Test Case 2 — Left Hemisphere Tumor

A smaller, well-defined tumor in the left hemisphere near the lateral ventricles, with a clearly demarcated hyperintense core. This case is more challenging from a path planning standpoint: the tumor's proximity to Broca's region requires a tightly constrained curved trajectory. The U-Net produced a compact, accurate mask without over-segmentation, and the path planner successfully navigated around the left-frontal region to reach the centroid from the inferior aspect.

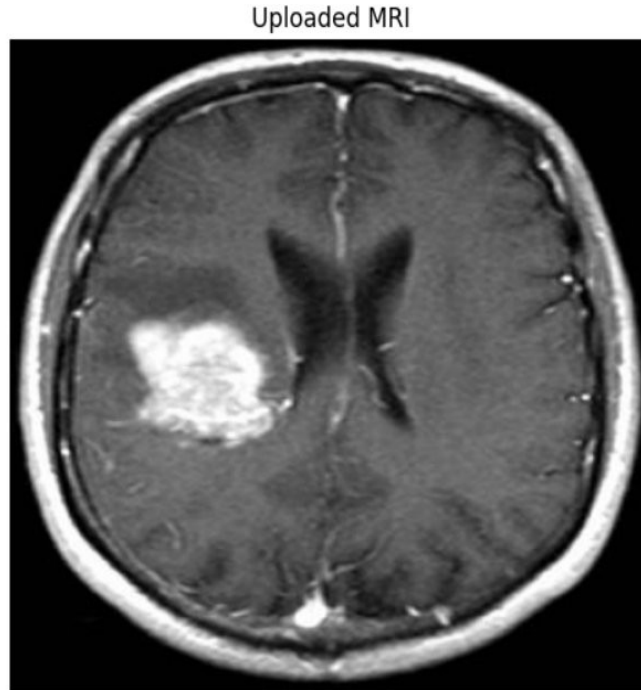


Figure 4: Case 2 — Original T1-weighted MRI showing a well-defined left-hemisphere tumor

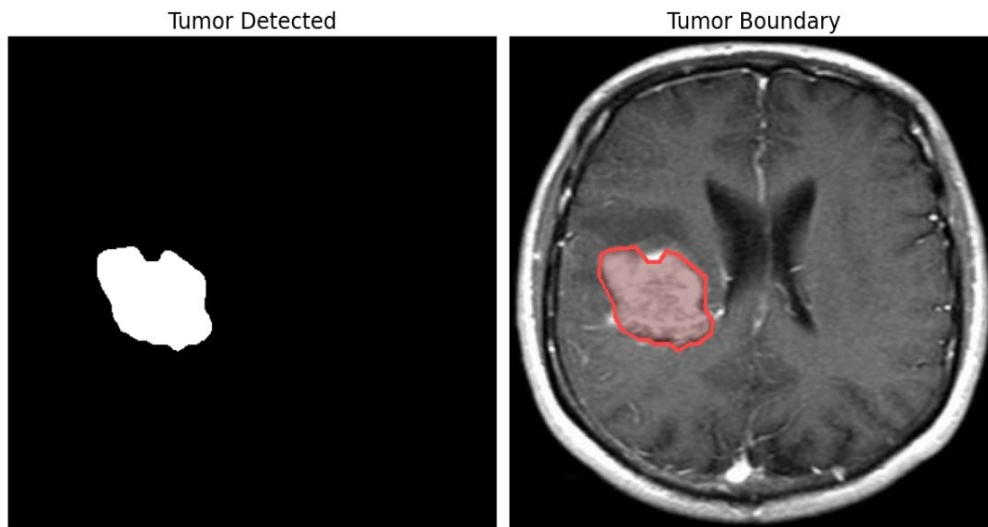


Figure 5: Case 2 — (Left) Binary segmentation mask | (Right) Tumor boundary overlaid on MRI

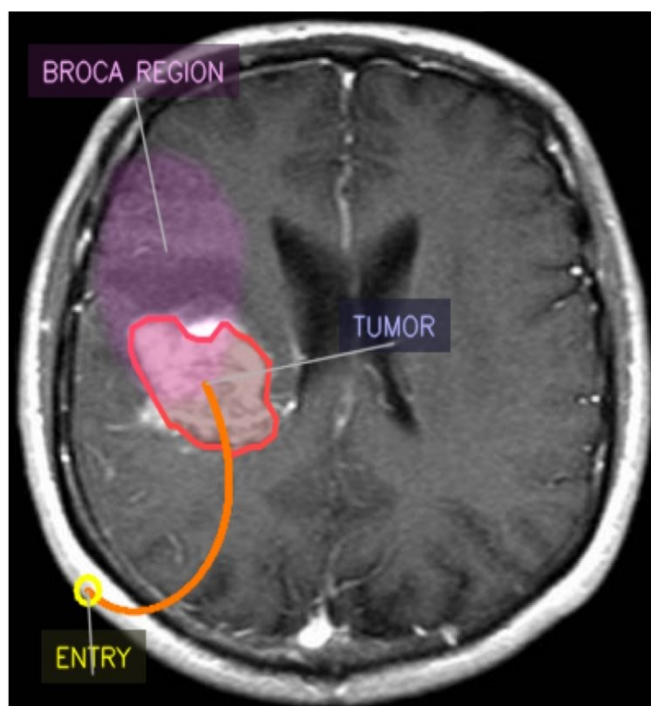


Figure 6: Case 2 — Probe path (orange) arcing below Broca's region (purple) before reaching the tumor centroid

7. Limitations

7.1 Training Data Volume

With 3,929 training images, the model is susceptible to domain shift when applied to scans from different scanners or acquisition protocols — reflected in the drop from 100% training accuracy to 76.7% on unseen data.

7.2 Fixed Anatomical Modelling

Broca's region is modelled as a fixed proportional polygon rather than being derived from patient-specific functional MRI or atlas registration. In clinical practice, its location varies across patients and must be individually identified.

7.3 Hardcoded Entry Point

The skull entry point is fixed at coordinates (12% width, 82% height). A clinical system would dynamically compute the optimal entry based on tumor location, vascular anatomy, and neurosurgical insertion angle constraints.

7.4 2D Operation Only

The pipeline operates on a single axial MRI slice. Clinical SLA requires 3D volumetric planning across the full MRI volume to define the true spatial extent of the tumor and the three-dimensional probe trajectory.

7.5 Computational Constraints

The prototype is limited by 8 GB RAM and Google Colab GPU memory, restricting model complexity and image resolution. Dedicated GPU infrastructure would remove these bottlenecks.

8. Future Work

- Expand training data using BraTS and TCIA datasets to improve cross-scanner generalisation.
- Replace the fixed Broca's polygon with atlas-based or fMRI-guided patient-specific localisation.
- Implement dynamic entry point optimisation accounting for vascular anatomy and insertion constraints.
- Extend the U-Net to a 3D architecture for full volumetric tumor segmentation and path planning.
- Add avoidance of additional eloquent regions: primary motor cortex, visual cortex (V1), and corticospinal tract.
- Conduct a clinical validation study comparing pipeline-planned trajectories against expert neurosurgeon plans.

9. Conclusion

This project presents a fully functional, end-to-end AI pipeline for stereotactic laser ablation probe path planning. Three components — MedGemma for detection, a fine-tuned U-Net for segmentation, and a Bézier path planner with Broca's region avoidance — are integrated into a seamless, Colab-executable workflow. The system achieved 93.33% detection accuracy, 76.7% segmentation accuracy on independent unseen data, and 100% path planning success across all tested cases.

The two test cases demonstrate the pipeline's effectiveness across different tumor locations and sizes, including the clinically demanding scenario where the tumor is proximal to Broca's region. While not yet ready for clinical deployment — owing to its fixed anatomical modelling, hardcoded entry point, and 2D-only operation — the prototype establishes a clear and extensible foundation for a clinically viable surgical planning tool. With expanded training data, patient-specific registration, and volumetric extension, this

approach has the potential to meaningfully lower the barrier to adoption of stereotactic laser ablation and extend its benefits to patients who currently have no surgical options.

10. References

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 Disclaimer: This is a research prototype not intended for clinical use. All outputs must be reviewed and validated by a qualified neurosurgeon before any clinical application.